

Homework Set HW12 due 4/1/04 at 11:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)

This is problem 2 Section 5.2 page 288.

Evaluate the sum using the summations formulas

$$\sum_{k=1}^{250} 2 = \underline{\hspace{2cm}}$$

2.(1 pt)

This is problem 4 Section 5.2 page 288.

Evaluate the sum using the summations formulas

$$\sum_{k=1}^{10} (k+1) = \underline{\hspace{2cm}}$$

3.(1 pt)

This is problem 8 Section 5.2 page 288.

Evaluate the sum using the summations formulas

$$\sum_{k=1}^{100} (k-1)^2 = \underline{\hspace{2cm}}$$

4.(1 pt)

This is problem 17 Section 5.3 page 301.

Evaluate the definite integral using linearity and subdivision together with the following results:

$$\int_{-1}^2 x^2 dx = 3; \quad \int_{-1}^0 x^2 dx = \frac{1}{3}; \quad \int_{-1}^2 x dx = \frac{3}{2}; \quad \int_0^2 x dx = 2$$

$$\int_0^{-1} x^2 dx = \underline{\hspace{2cm}}$$

5.(1 pt)

This is problem 18 Section 5.3 page 301.

Evaluate the definite integral using linearity and subdivision together with the following results:

$$\int_{-1}^2 x^2 dx = 3; \quad \int_{-1}^0 x^2 dx = \frac{1}{3}; \quad \int_{-1}^2 x dx = \frac{3}{2}; \quad \int_0^2 x dx = 2$$

$$\int_{-1}^2 (x^2 + x) dx = \underline{\hspace{2cm}}$$

6.(1 pt)

This is problem 20 Section 5.3 page 301.

Evaluate the definite integral using linearity and subdivision together with the following results:

$$\int_{-1}^2 x^2 dx = 3; \quad \int_{-1}^0 x^2 dx = \frac{1}{3}; \quad \int_{-1}^2 x dx = \frac{3}{2}; \quad \int_0^2 x dx = 2$$

$$\int_0^2 x^2 dx = \underline{\hspace{2cm}}$$

7.(1 pt)

This is problem 22 Section 5.3 page 301.

Evaluate the definite integral using linearity and subdivision together with the following results:

$$\int_{-1}^2 x^2 dx = 3; \quad \int_{-1}^0 x^2 dx = \frac{1}{3}; \quad \int_{-1}^2 x dx = \frac{3}{2}; \quad \int_0^2 x dx = 2$$

$$\int_{-1}^0 (3x^2 - 5x) dx = \underline{\hspace{2cm}}$$